



# 3D Human Motion Capture and Generation

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# Outline

- Introduction
- Background & Motivation
- Proposed Methods & Performance Evaluation
  - 3D Human Motion Capture from Videos
  - 3D Human Motion Generation from Text Inputs
  - 3D Human Animation
- Conclusions
- Q & A





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# Introduction – 3D Human Motion (1/5)

- 3D human motion is useful for
  - **Video generation**
  - Animation production
  - Pedestrian analysis



Figure 1: Video Generation







# Introduction – 3D Human Motion (2/5)

- 3D human motion is useful for
  - Video generation
  - **Animation production**
  - Pedestrian analysis



Figure 2: Animation production





# Introduction – 3D Human Motion (3/5)

- 3D human motion is useful for
  - Video generation
  - Animation production
  - **Pedestrian analysis**



Figure 3: Pedestrian analysis





# Introduction – 3D Human Motion (4/5)

- 3D human motion representation



Outdoor video

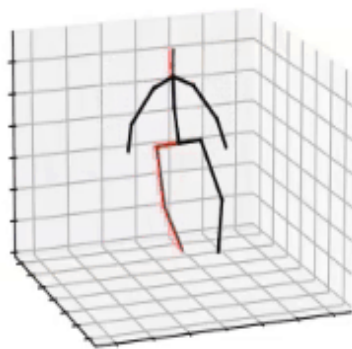
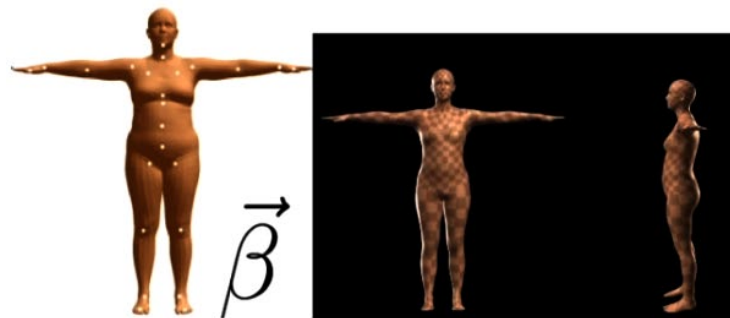


Figure 4: 3D keypoints

Shape: PCA coefficients



Pose: Rotation of 23 joints

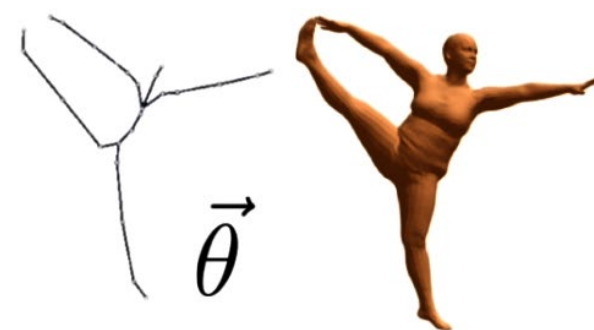


Figure 5: Parametric models: SMPL/SMPL-X







# Introduction – 3D Human Motion (5/5)

- Challenges for traditional motion capture

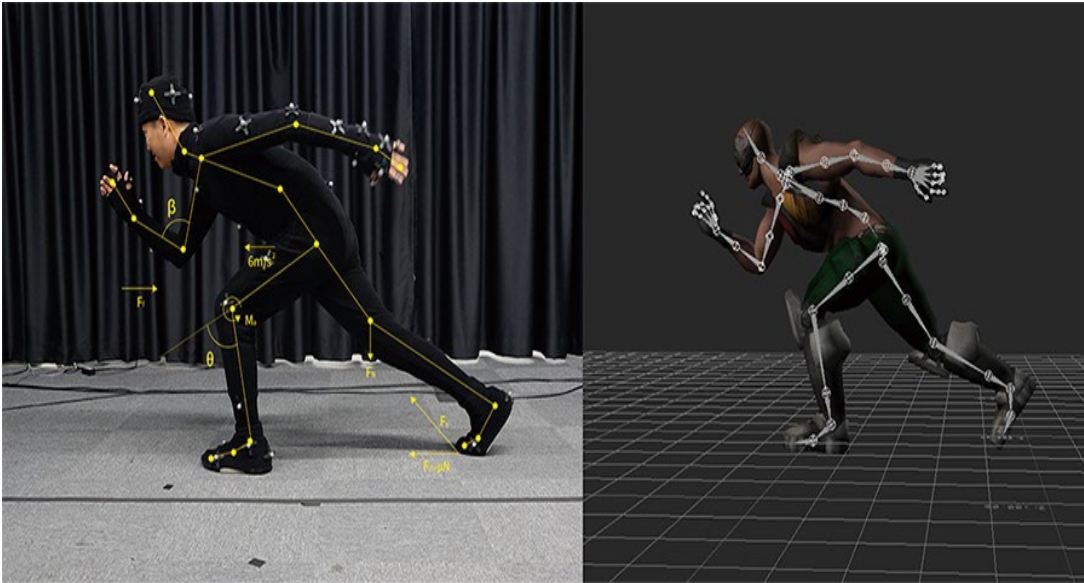


Figure 6: High-cost



Figure 7: Hard to scale





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# Background & Motivation (1/6)

- Easy inputs
  - Images
  - Videos
  - Text prompts



Figure 8: Images





# Background & Motivation (2/6)

- Easy inputs
  - Images
  - **Videos**
  - Text prompts

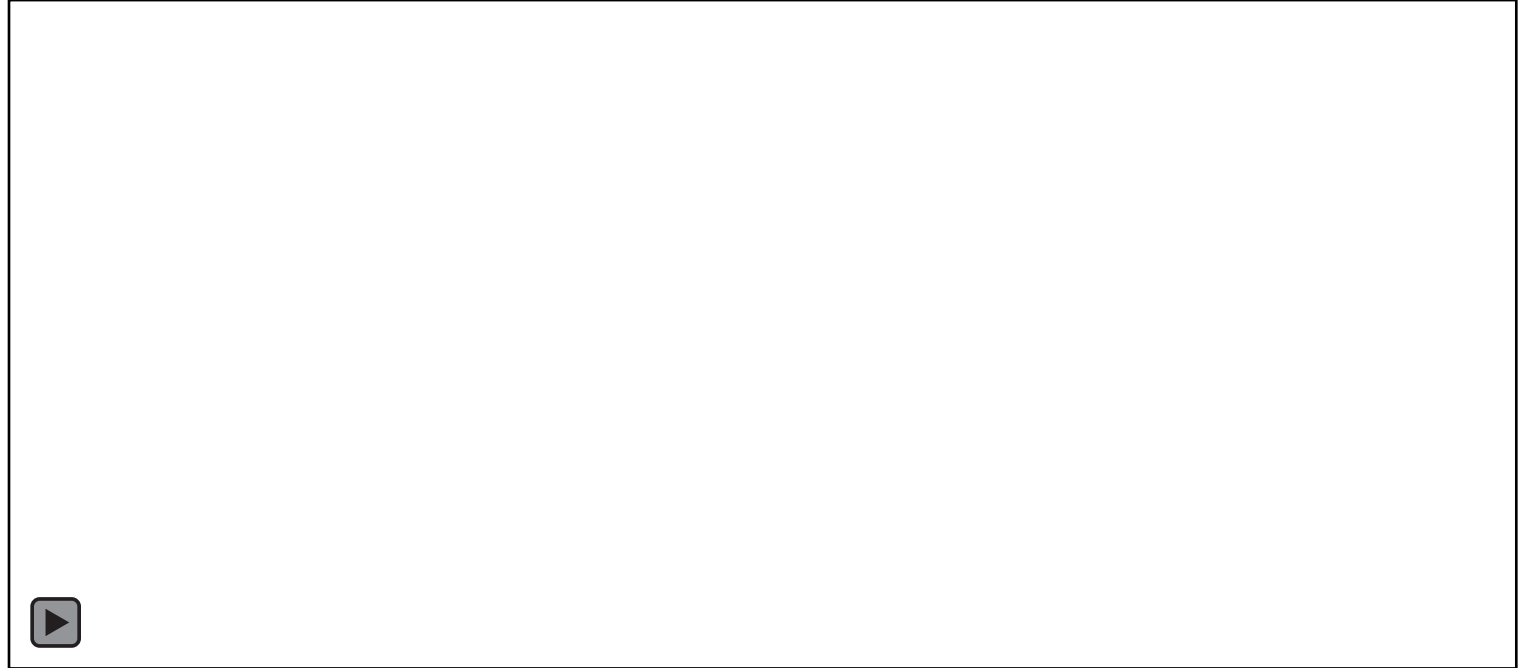


Figure 9:Videos





# Background & Motivation (3/6)

- Easy inputs
  - Images
  - Videos
  - **Text prompts**

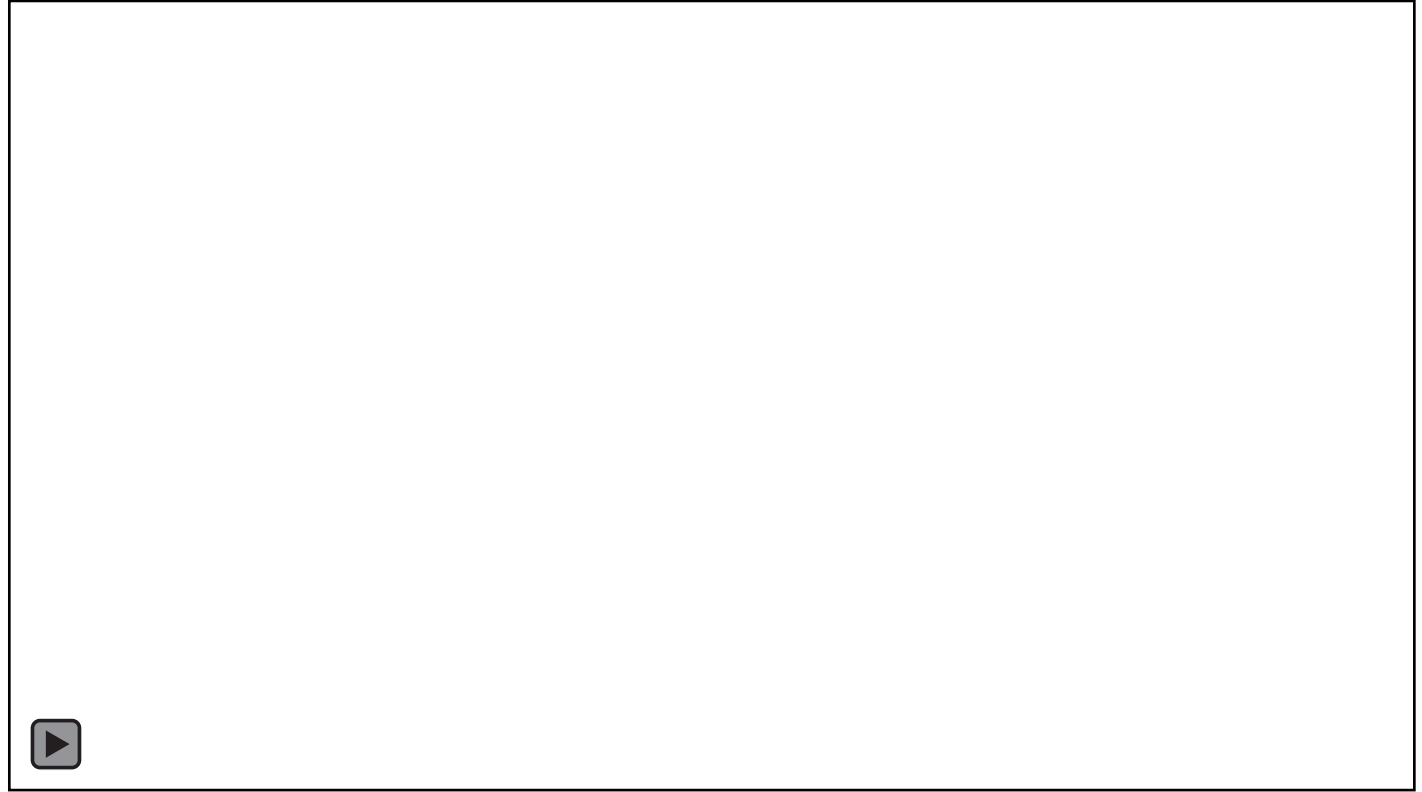


Figure 10: Text Prompts







# Background & Motivation (4/6)

- Motivation
- Limitations of cascaded systems:
  - Accumulated errors
  - Temporal information is not fully exploited
  - End to end training is not supported

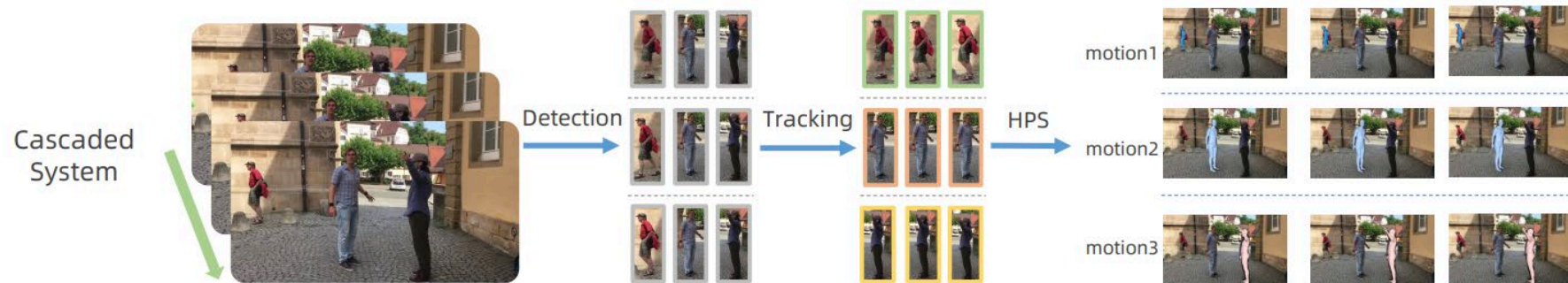


Figure 11: Motivation for Motion capture from video





# Background & Motivation (5/6)

- Motivation
  - Correspondence between text and motion may be weak
  - The mutual property of T2M and M2T is usually overlooked

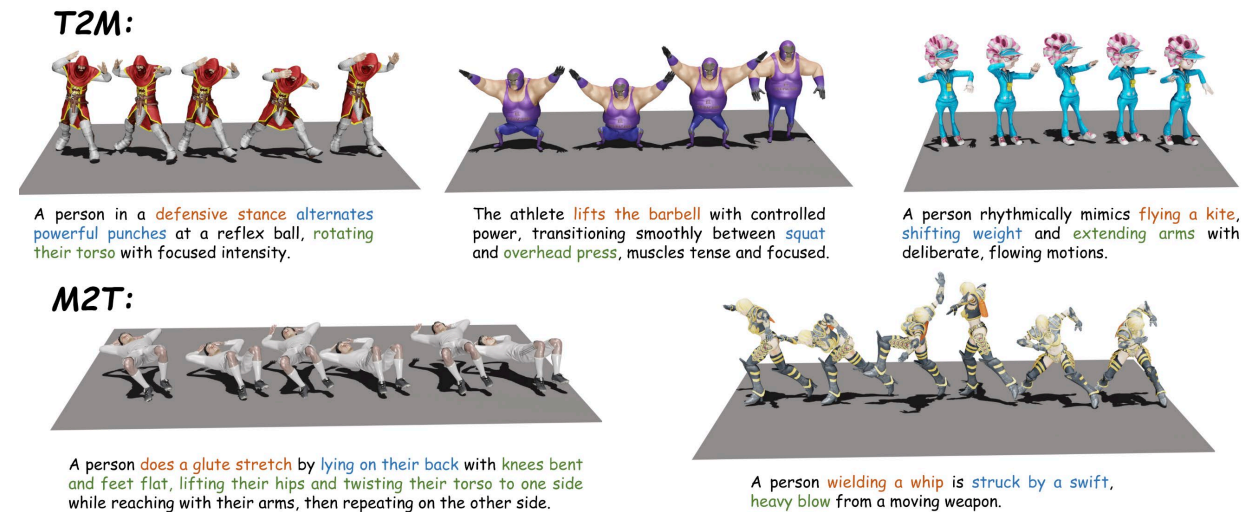


Figure 12: Motivation for 3D Human Motion Generation from Text Inputs





# Background & Motivation (6/6)

- Motivations
  - Inconsistent setting for human NeRFs: Train & test view and frame split, dataset split
  - Accurate SMPL parameters may not always available
  - Generalizable human NeRF from monocular videos is not explored

Table 1: Motivation for 3D Human Animation

| Method             | Dataset          | Views      | Frames                    | Estimated | Generalizable | Animatable | Unified Evaluation |
|--------------------|------------------|------------|---------------------------|-----------|---------------|------------|--------------------|
| NeuralBody [10]    | ZM, PS           | 4          | 100-300                   | ✗         | ✗             | ✓          | ✗                  |
| AniNeRF [197]      | ZM, H36M         | 4          | 100-300                   | ✗         | ✗             | ✓          | ✗                  |
| Arah [204]         | ZM, H36M         | 4          | 300-400                   | ✗         | ✗             | ✓          | ✗                  |
| HumanNeRF [47]     | ZM               | 1          | 500-600                   | ✗         | ✗             | ✓          | ✗                  |
| UV-Volume [196]    | ZM, H36M         | 18         | 100                       | ✗         | ✗             | ✓          | ✗                  |
| MonoHuman [195]    | ZM               | 1          | 500-600                   | ✗         | ✗             | ✓          | ✗                  |
| NHP [200]          | ZM, AIST++       | 3*         | 1* or 3*                  | ✗         | ✓             | ✗          | ✗                  |
| MPSNeRF [201]      | ZM, H36M, THuman | 3*         | 1*                        | ✗         | ✓             | ✓          | ✗                  |
| GP-NeRF [199]      | ZM               | 3*         | 1*                        | ✗         | ✓             | ✗          | ✗                  |
| KeypointNeRF [198] | ZM               | 3*         | 1*                        | ✗         | ✓             | ✗          | ✗                  |
| GNR [12]           | ZM, GB, RP       | 4*         | 1*                        | ✗         | ✓             | ✗          | ✗                  |
| Ours               | ZM, GB, HM       | 1 or 4, 1* | 1, 60, 100, 300, 500, 10* | ✓         | ✓             | ✓          | ✓                  |





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# Motion Capture from Videos (1/11)

- Input: a monocular video
- Output: 3D human motion representation (i.e. SMPL/SMPL-X)

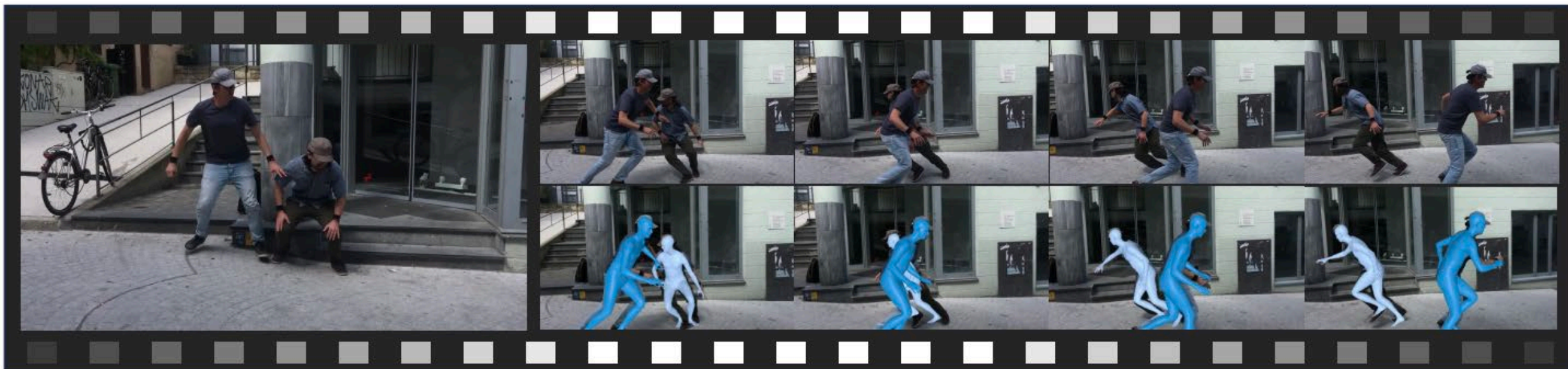


Figure 13: From a monocular video to motion representation





# Motion Capture from Videos (2/11)

- Method
- Key idea
  - End-to-end
  - One query responsible for one person
  - Coarse to fine

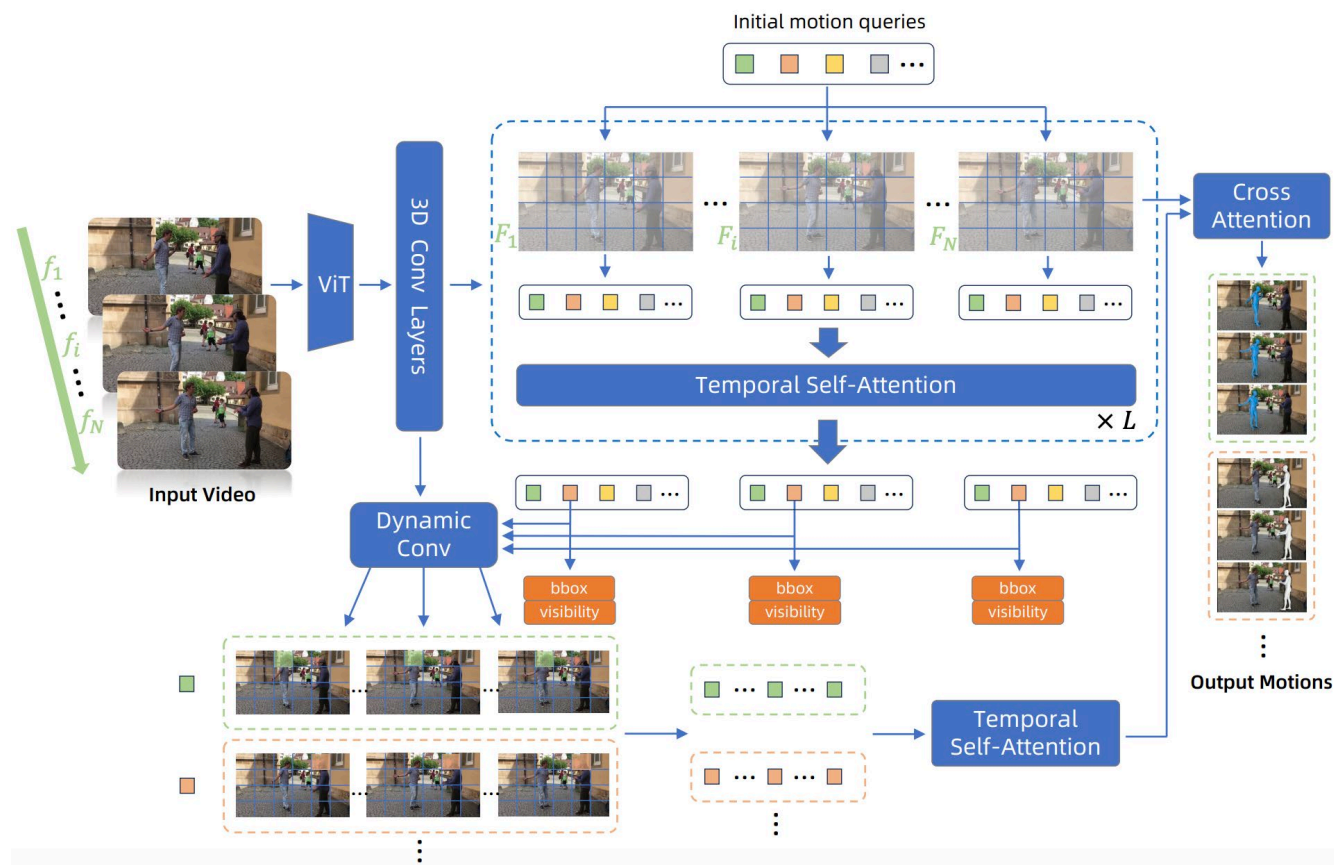


Figure 14: Method for for Motion capture from vedio







# Motion Capture from Videos (3/11)

- A new dataset
  - **Synthetic**
  - Accurate
  - Diverse
  - Low-cost and Scalable



Figure 15: Synthetic Property of the M3C Dataset





# Motion Capture from Videos (4/11)

- A new dataset
- Synthetic
- **Accurate**
- Diverse
- Low-cost and Scalable

Table 2: Accurate Property of the M3C Dataset

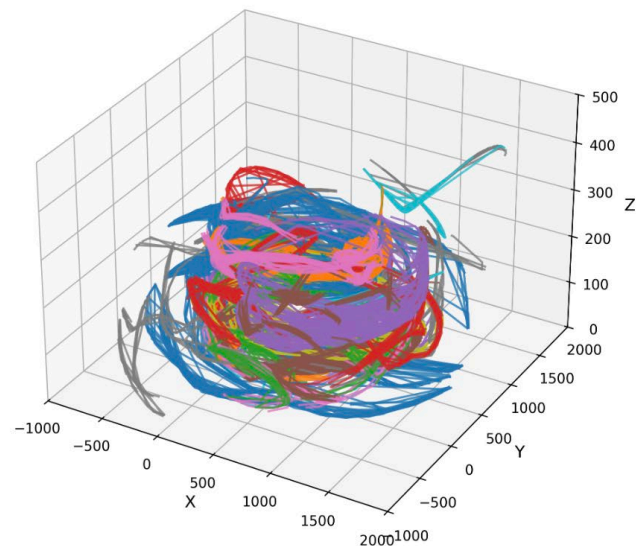
| Dataset                    | Clips | Frames | Moving Camera | Multi Person | Annotation Type | Camera Traj  | Scene type     |
|----------------------------|-------|--------|---------------|--------------|-----------------|--------------|----------------|
| PoseTrack [2]              | 514   | 66K    | Yes           | Yes          | 2D KPT          | -            | Outdoor        |
| MuPoTS-3D [36]             | 20    | 8K     | Yes           | Yes          | 3D KPT          | -            | Outdoor        |
| 3DPW [48]                  | 60    | 51K    | No            | Yes          | SMPL            | -            | Outdoor        |
| Humanvid [51] (Synthetic)  | 100K  | 12M    | Yes           | No           | SMPL-X          | Ground Truth | Outdoor        |
| BEDLAM [5] (Static Camera) | 8.7K  | 1.35M  | No            | Yes          | SMPL-X          | -            | Outdoor/Indoor |
| BEDLAM [5] (Moving Camera) | 1.75K | 250K   | Yes           | Yes          | SMPL-X          | Ground Truth | Indoor         |
| M3C (ours)                 | 6K    | 1M     | Yes           | Yes          | SMPL-X          | Ground Truth | Outdoor        |



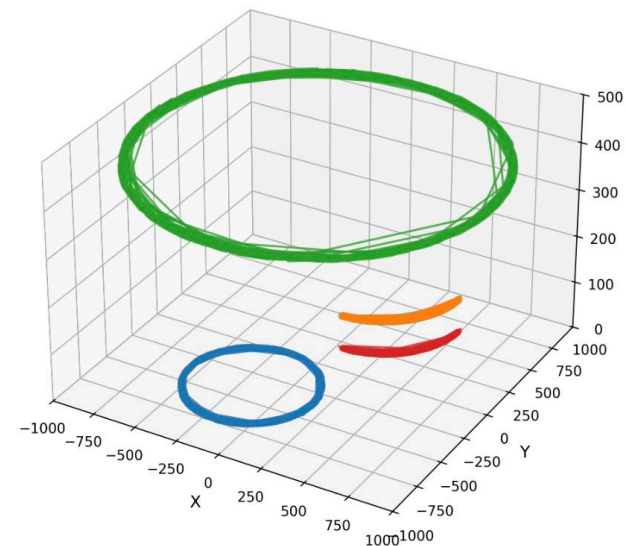


# Motion Capture from Videos (5/11)

- A new dataset
  - Synthetic
  - Accurate
  - **Diverse**
  - Low-cost and Scalable



M3C Camera Trajectories



BEDLAM Camera Trajectories

Figure 16: Accurate Property of the M3C Dataset







# Motion Capture from Videos (6/11)

- A new dataset
  - Synthetic
  - Accurate
  - **Diverse**
  - Low-cost and Scalable



Figure 17: Diverse Property of the M3C Dataset







# Motion Capture from Videos (7/11)

- A new dataset
  - Synthetic
  - Accurate
  - Diverse
  - **Low-cost and Scalable**



Figure 18: Low-cost and Scalable Property of the M3C Dataset





# Motion Capture from Videos (8/11)

## Experiments

Table 3: Multi-Person Tracking Results on Real Data

Real

| Methods                   | IDs ↓    | MOTA ↑      | IDF1 ↑      | HOTA ↑      |
|---------------------------|----------|-------------|-------------|-------------|
| PHALP [39]                | 5        | 97.9        | 93.7        | 74.2        |
| YOLOX [14]+ByteTrack [54] | 3        | 97.3        | 98.2        | 73.1        |
| BEV [46]+ByteTrack [54]   | 37       | 93.6        | 79.1        | 59.3        |
| TRACE [47]                | 1        | 99.3        | 99.7        | <b>74.7</b> |
| WHAM [43]                 | 59       | 97.8        | 68.2        | 59.9        |
| Ours                      | <b>0</b> | <b>99.6</b> | <b>99.8</b> | 70.8        |

Table 4: 3D Pose Estimation Results on Real Data

|                            | Methods       | PAMPJPE ↓   | MPJPE ↓     | PVE ↓       |
|----------------------------|---------------|-------------|-------------|-------------|
| Single-frame<br>(w/ GT BB) | HybrIK [24]   | 45.0        | 74.1        | 86.5        |
|                            | METRO [29]    | 47.9        | 77.1        | 88.2        |
|                            | GLAMR [53]    | 50.7        | -           | -           |
|                            | CLIFF [26]    | 43.0        | 69.0        | 81.2        |
|                            | D&D [25]      | 42.7        | 73.7        | 88.6        |
|                            | OSX [27]      | 45.1        | 74.7        | -           |
|                            | Multi-HMR [3] | 43.8        | 64.6        | 79.7        |
| Video<br>(w/o GT BB)       | ROMP [45]     | 47.3        | 76.7        | 93.4        |
|                            | BEV [46]      | 46.9        | 78.5        | 92.3        |
|                            | TRACE [47]    | 50.8        | 80.3        | 98.1        |
|                            | Ours          | <b>44.7</b> | <b>72.6</b> | <b>84.9</b> |

Table 5: Comprehensive Comparison on Synthetic Data

Synthetic

| Methods                                        | IDs ↓      | MOTA ↑       | IDF1 ↑       | HOTA ↑       | MPJPE ↓      | PAMPJPE ↓    | PVE ↓        | Inference Time ↓ |
|------------------------------------------------|------------|--------------|--------------|--------------|--------------|--------------|--------------|------------------|
| Fast-RCNN [15] + SMPLer-X [7] + ByteTrack [54] | 1660       | 78.83        | 81.09        | 70.41        | 119.19       | 52.62        | 129.22       | 0.463            |
| Multi-HMR [3] + ByteTrack [54]                 | 515        | 91.35        | 92.54        | 84.59        | 81.58        | <b>42.93</b> | 88.88        | 0.201            |
| Ours                                           | <b>128</b> | <b>95.15</b> | <b>97.01</b> | <b>97.73</b> | <b>79.88</b> | 45.56        | <b>86.40</b> | <b>0.027</b>     |







# Motion Capture from Videos (9/11)

- Experiments

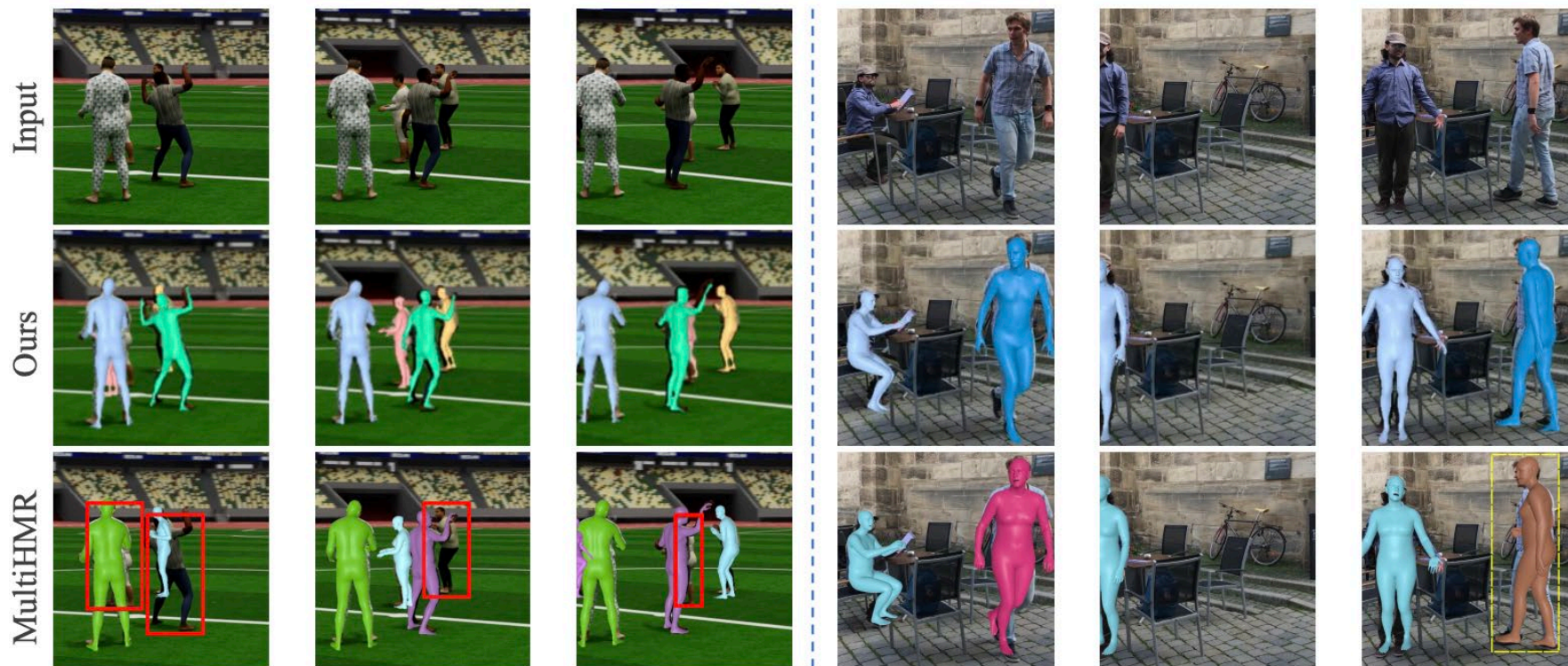


Figure 19: Experiment-1 for Capture from videos





# Motion Capture from Videos (10/11)

- Experiments

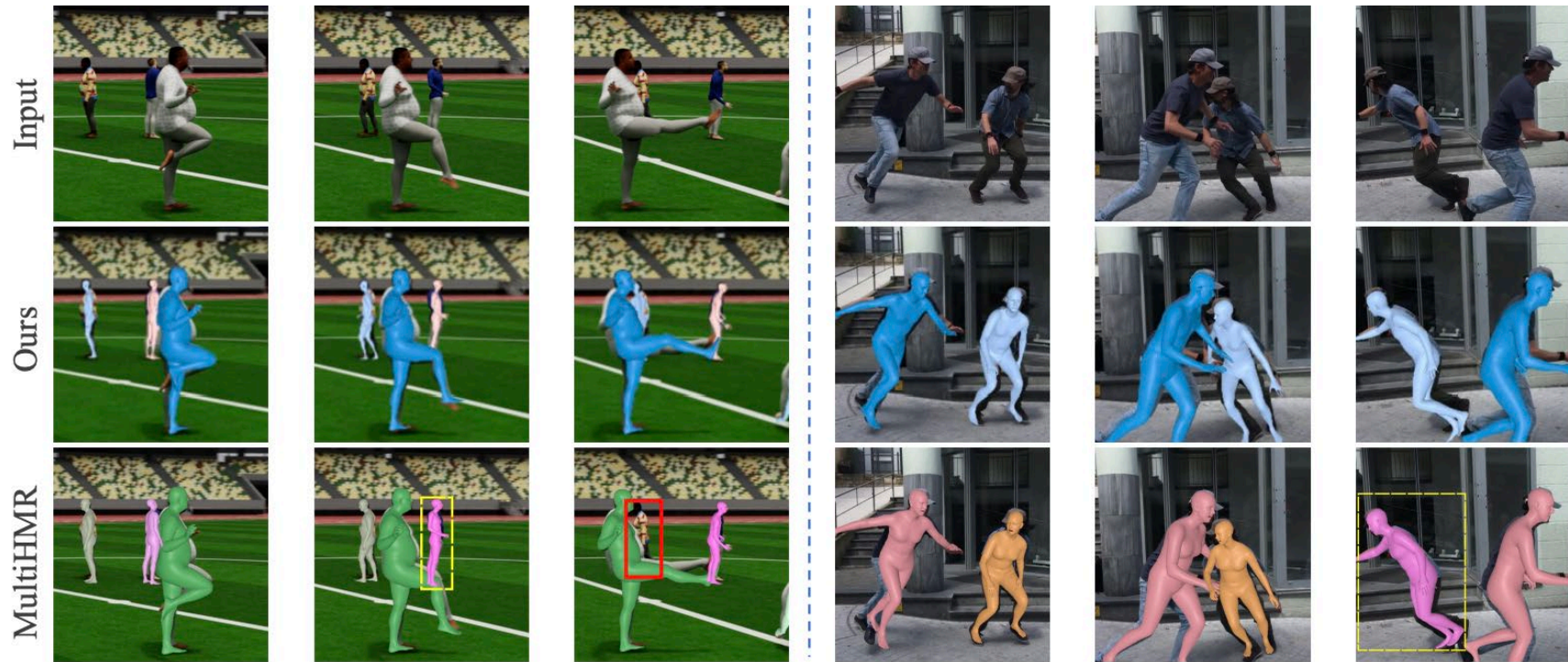


Figure 20: Experiment-2 for Capture from videos







# Motion Capture from Videos (11/11)

- Experiments



Figure 21: Experiment-3 for Capture from videos







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# Motion Generation from Text (1/10)

- Input: text prompts
- Output: 3D human motion representation (i.e. SMPL/SMPL-X)

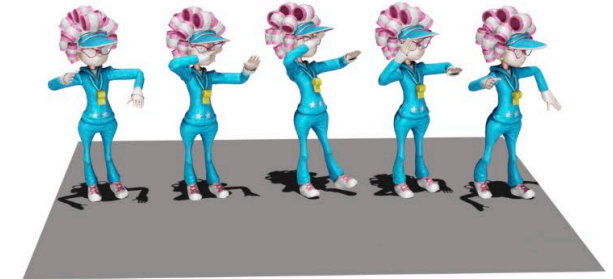
**T2M:**



A person in a **defensive stance** **alternates powerful punches** at a reflex ball, **rotating their torso** with focused intensity.



The athlete **lifts the barbell** with controlled power, transitioning smoothly between **squat** and **overhead press**, muscles tense and focused.



A person rhythmically mimics **flying a kite**, **shifting weight** and **extending arms** with deliberate, flowing motions.

Figure 22: From text prompts to 3D motion Representation





# Motion Generation from Text 2/10)

- Method
- Key design
- CoT as semantic bridge → Bidirectional training → Two-stage training with RL

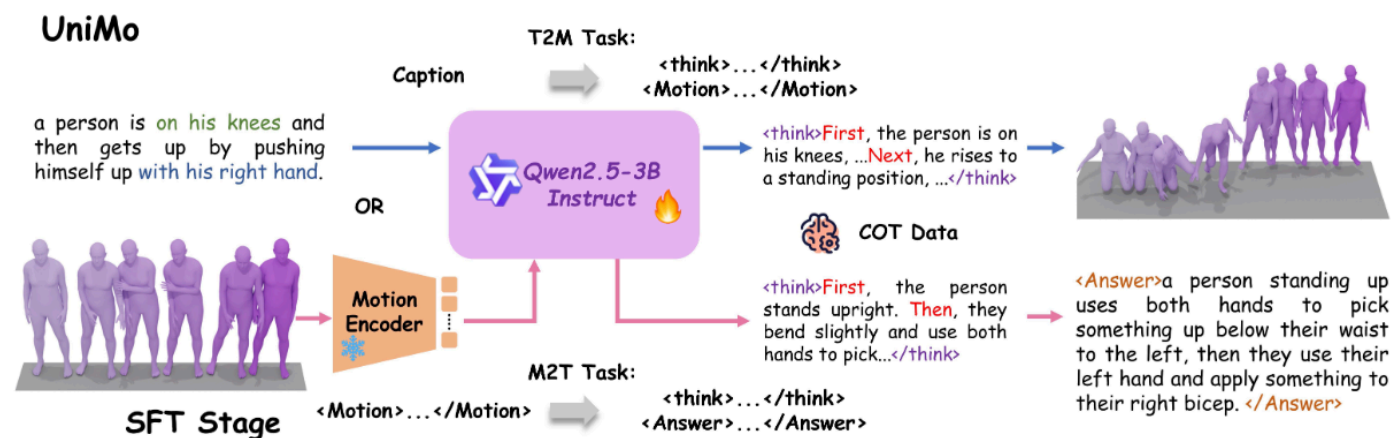


Figure 23: Method for 3D Human Motion Generation from Text Inputs





# Motion Generation from Text (3/10)

- Method
- Key design

1. CoT as semantic bridge.

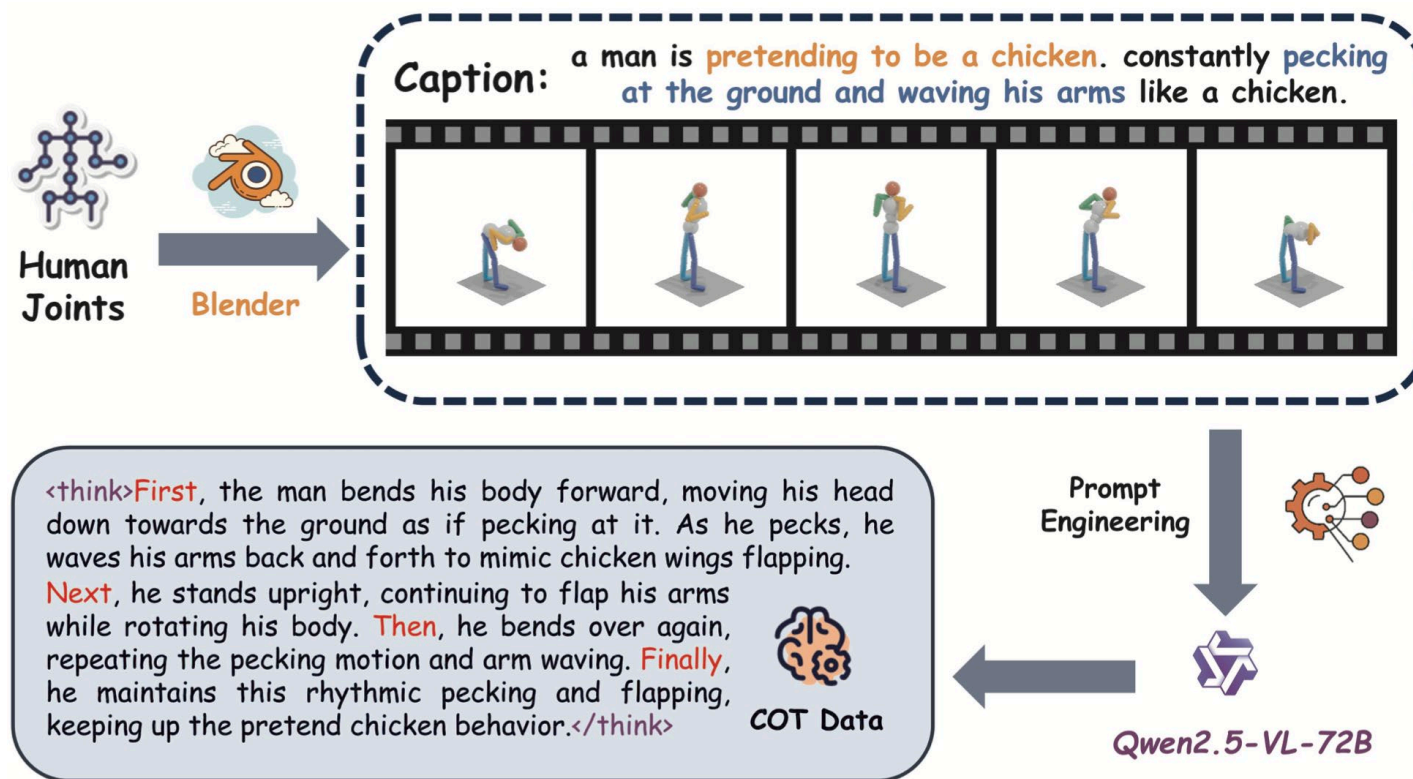


Figure 24: Step 1 for 3D Human Motion Generation from Text Inputs







# Motion Generation from Text (4/10)

- Method
- Key design

## 2. Bidirectional training: Motion to text and text to motion

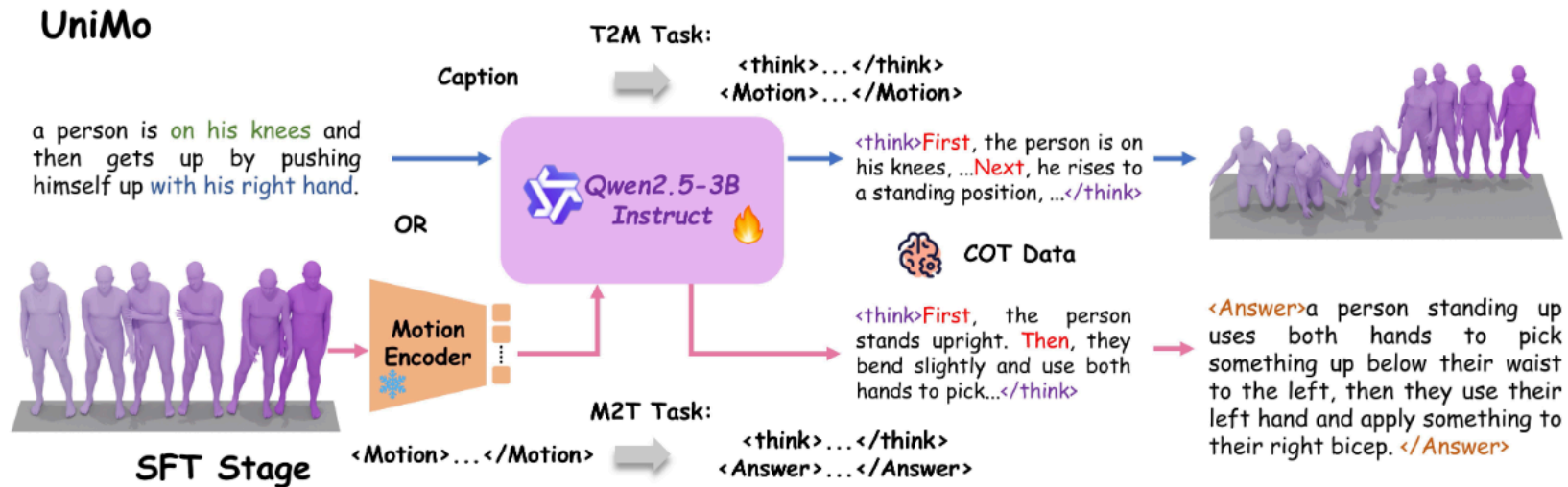


Figure 25: Step 2 for 3D Human Motion Generation from Text Inputs





# Motion Generation from Text (5/10)

- Method
- Key design

## 3. Two-stage training with RL

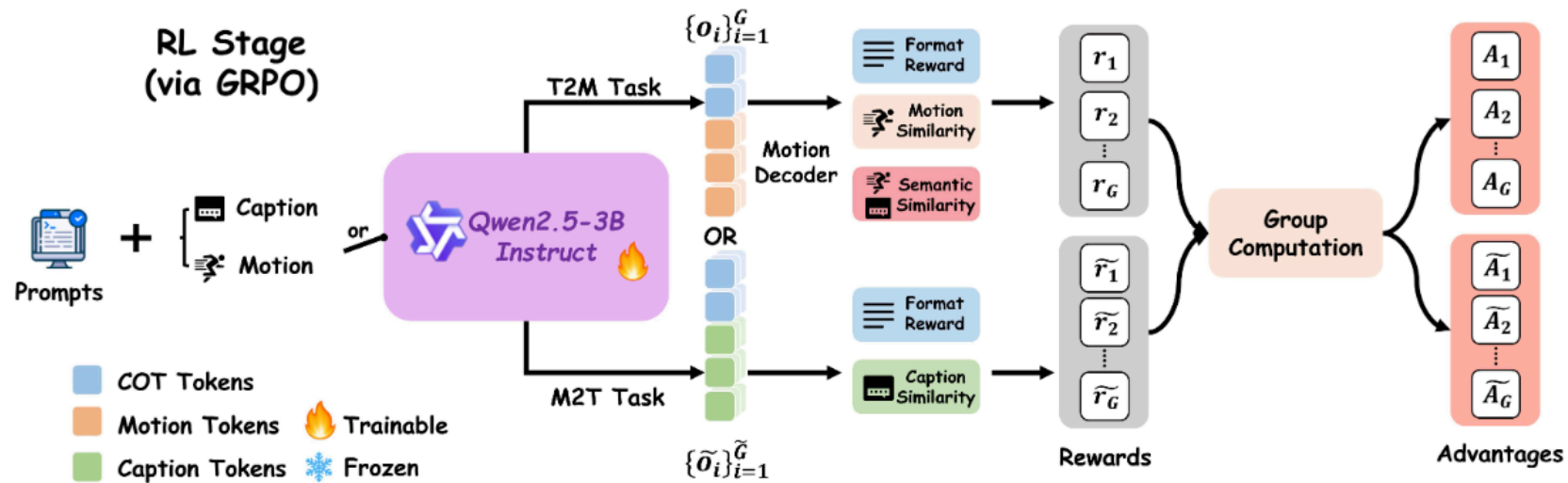


Figure 26: Step 3 for 3D Human Motion Generation from Text Inputs





# Motion Generation from Text (6/10)

- Experiments
- Text to motion benchmark

Table 6: Experiment for 3D Human Motion Generation from Text Inputs

| Methods            | R-Precision $\uparrow$   |                          |                          | FID $\downarrow$         | MM-Dist $\downarrow$     | Diversity $\uparrow$      | MModality $\uparrow$     |
|--------------------|--------------------------|--------------------------|--------------------------|--------------------------|--------------------------|---------------------------|--------------------------|
|                    | Top1                     | Top2                     | Top3                     |                          |                          |                           |                          |
| MDM [146]          | 0.320 $\pm$ 0.005        | 0.498 $\pm$ 0.004        | 0.611 $\pm$ 0.007        | 0.544 $\pm$ 0.044        | 5.566 $\pm$ 0.027        | 9.559 $\pm$ 0.086         | <b>2.799</b> $\pm$ 0.074 |
| MLD [43]           | 0.481 $\pm$ 0.003        | 0.673 $\pm$ 0.003        | 0.772 $\pm$ 0.002        | 0.473 $\pm$ 0.013        | 3.196 $\pm$ 0.010        | 9.724 $\pm$ 0.082         | 2.413 $\pm$ 0.072        |
| MotionDiffuse [44] | 0.491 $\pm$ 0.001        | 0.681 $\pm$ 0.001        | 0.782 $\pm$ 0.001        | 0.630 $\pm$ 0.001        | 3.113 $\pm$ 0.001        | 9.410 $\pm$ 0.049         | 1.553 $\pm$ 0.064        |
| T2M [139]          | 0.457 $\pm$ 0.002        | 0.559 $\pm$ 0.007        | 0.740 $\pm$ 0.003        | 1.067 $\pm$ 0.002        | 3.340 $\pm$ 0.008        | 9.188 $\pm$ 0.002         | 2.090 $\pm$ 0.088        |
| TM2T [145]         | 0.424 $\pm$ 0.003        | 0.618 $\pm$ 0.003        | 0.729 $\pm$ 0.002        | 1.501 $\pm$ 0.017        | 3.467 $\pm$ 0.011        | 8.589 $\pm$ 0.076         | <u>2.424</u> $\pm$ 0.079 |
| T2M-GPT [152]      | 0.491 $\pm$ 0.003        | 0.680 $\pm$ 0.003        | 0.775 $\pm$ 0.002        | <u>0.116</u> $\pm$ 0.004 | 3.118 $\pm$ 0.011        | 9.761 $\pm$ 0.081         | 1.856 $\pm$ 0.111        |
| MotionGPT [9]      | 0.492 $\pm$ 0.003        | 0.681 $\pm$ 0.003        | 0.778 $\pm$ 0.002        | 0.232 $\pm$ 0.008        | 3.096 $\pm$ 0.008        | 9.528 $\pm$ 0.071         | 2.008 $\pm$ 0.083        |
| MoMask [7]         | <u>0.521</u> $\pm$ 0.002 | 0.713 $\pm$ 0.002        | 0.807 $\pm$ 0.002        | <b>0.045</b> $\pm$ 0.002 | 2.958 $\pm$ 0.008        | 9.620 $\pm$ 0.064         | 1.241 $\pm$ 0.064        |
| MotionChain [153]  | 0.504 $\pm$ 0.003        | 0.617 $\pm$ 0.002        | 0.790 $\pm$ 0.003        | 0.248 $\pm$ 0.009        | 3.033 $\pm$ 0.010        | 9.470 $\pm$ 0.075         | 1.727 $\pm$ 0.014        |
| MotionLLM [8]      | 0.515 $\pm$ 0.004        | 0.691 $\pm$ 0.003        | 0.801 $\pm$ 0.004        | 0.230 $\pm$ 0.009        | 2.967 $\pm$ 0.020        | 9.908 $\pm$ 0.102         | 2.142 $\pm$ 0.014        |
| MotionGPT-2 [137]  | 0.496 $\pm$ 0.002        | 0.691 $\pm$ 0.003        | 0.782 $\pm$ 0.004        | 0.191 $\pm$ 0.004        | 3.080 $\pm$ 0.013        | 9.860 $\pm$ 0.026         | 2.137 $\pm$ 0.022        |
| Motion-R1 [45]     | 0.515 $\pm$ 0.003        | <u>0.719</u> $\pm$ 0.002 | <u>0.818</u> $\pm$ 0.002 | 0.201 $\pm$ 0.004        | <u>2.854</u> $\pm$ 0.010 | <u>10.026</u> $\pm$ 0.075 | 2.317 $\pm$ 0.105        |
| UniMo (Ours)       | <b>0.539</b> $\pm$ 0.003 | <b>0.738</b> $\pm$ 0.002 | <b>0.831</b> $\pm$ 0.002 | 0.177 $\pm$ 0.004        | <b>2.768</b> $\pm$ 0.010 | <b>10.042</b> $\pm$ 0.076 | 1.924 $\pm$ 0.080        |





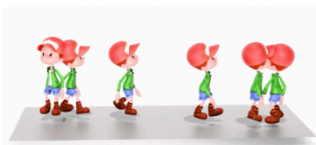
# Motion Generation from Text (7/10)

- Experiments
- Text to motion benchmark

## T2M Task:

Person turns around so their back is facing the front. They then walk forward for a moment before sliding to the right.

### MoMask



### MotionLLM



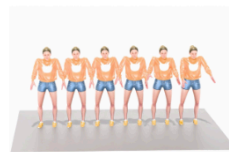
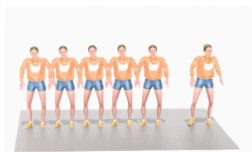
### CoT (Ours)

`<think>`First, the person turns around so their back is facing the front. As they complete this turn, they begin to walk forward for a moment. Next, they slide to the right, maintaining their forward momentum. Finally, they settle into a new position slightly to the right of their original spot.`</think>`

### Ours



A person shakes their head in disbelief, then looks around the room.



`<think>` First, the person shakes their head in a disoriented manner. Then, they turn their head from side to side as if searching for something or someone. Meanwhile, they remain standing in place, maintaining a neutral stance throughout the sequence.`</think>`



A person jumps up excitedly, throwing their arms over the head in celebration.



`<think>`First, the person stands on a platform, with their arms at their sides. As they begin to jump up, they raise their arms above their head. Meanwhile, they maintain an excited posture, showing excitement. While in mid-jump, they keep their arms elevated. After that, they land back on the platform, lowering their arms slightly but still keeping a sense of energy. Finally, they return to a standing position, with their arms at their sides.`</think>`



Figure 27:prompts for Generation from Text







# Motion Generation from Text (8/10)

- Experiments
- Text to motion



Figure 28: Examples for Text to Motion





# Motion Generation from Text (9/10)

- Experiments
- Motion to text benchmark

Table 7: Experiment for Motion to text benchmark

| Captioning        | BLEU@1↑      | BLEU@4↑      | ROUGE-L↑    | CIDEr↑       | BertScore↑   |
|-------------------|--------------|--------------|-------------|--------------|--------------|
| TM2T [145]        | 48.90        | 8.27         | 38.1        | 15.80        | 32.2         |
| LaMPM2T [176]     | 47.8         | 13.04        | 37.1        | 28.9         | 32.7         |
| MoTe [177]        | 46.7         | 11.15        | 37.4        | 31.5         | 30.3         |
| MotionGPT [9]     | 48.20        | 12.47        | 37.4        | 29.20        | 32.4         |
| MotionGPT-2 [137] | 48.7         | 13.8         | 37.6        | 29.8         | 32.6         |
| MotionChain [153] | 48.10        | 12.56        | 33.9        | 33.70        | 36.9         |
| MotionLLM [8]     | <u>54.53</u> | <u>17.65</u> | <u>48.7</u> | <u>33.74</u> | <u>42.63</u> |
| UniMo (Ours)      | <b>63.10</b> | <b>19.74</b> | <b>48.8</b> | <b>46.69</b> | <b>54.26</b> |





# Motion Generation from Text (10/10)

- Experiments
- Motion to text

| M2T Task:                                                                           | MotionGPT                                                                                                     | MotionLLM                                                                                      | Ours (with CoT)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | GT                                                                                                                                                                         |
|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|   | a person is doing warm ups.                                                                                   | a person stretches their arms by crossing them over the back of their head.                    | <p>&lt;think&gt;First, the person lifts both arms in front of their body. As they reach the peak, they begin to lower their arms back down. Finally, the person returns their arms to their original position by their sides.&lt;/think&gt;<br/>&lt;Answer&gt;a person is putting their arms out in front of them and then putting them back down.&lt;/Answer&gt;</p>                                                                                                                                                                               | a man has his arms up to his chest as if to do a pushing motion and moves his shoulders back and forward to stretch out his chest and wiggles arms down to their position. |
|   | a person rotates their right hand and arm in a clockwise motion in front of themselves parallel to the floor. | a person moves their right hand in a circular motion in front of them.                         | <p>&lt;think&gt;First, the person stands still with their right arm slightly bent. As they begin to stir, they move their right hand in a circular motion, seemingly stirring something unseen. Meanwhile, their left arm remains stationary. Next, they continue this stirring action, maintaining a consistent circular movement with their right hand. Finally, they keep the stirring motion going, suggesting a focused effort on the task.&lt;/think&gt;<br/>&lt;Answer&gt;a person stirs something with their right hand.&lt;/Answer&gt;</p> | a man moves his right arm around in a clockwise circle a few times. his palm is facing down as though he is stirring something.                                            |
|  | a person slowly walked forward and grabbed something and put it somewhere                                     | a person walks forward, squats down, stands back up, turns around, and then squats down again. | <p>&lt;think&gt;First, the person walks forward. Then, they crouch down while facing forward. As they finish crouching, they turn around and walk back. Next, they crouch down again in the opposite direction. Finally, the person stands up after completing the crouch.&lt;/think&gt;<br/>&lt;Answer&gt;a person picks something up and moves it to their right side and then puts it back down.&lt;/Answer&gt;</p>                                                                                                                              | a person walks forward, then leans down as if to pick something up, turns around and puts something down on the ground.                                                    |

Figure 29:Experiment for Motion to Text





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- Introduction
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# 3D Human Animation(1/5)

- Input: 3D human motions
- Output: animated videos

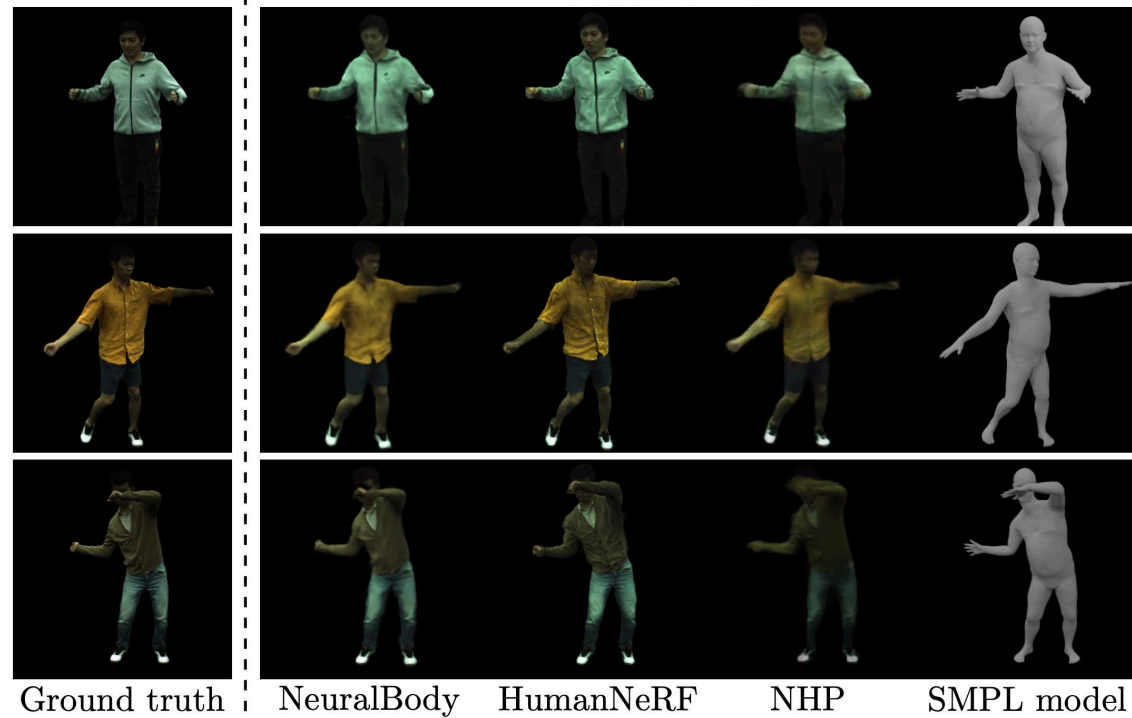


Figure 30: From 3D human motions to animated videos





# 3D Human Animation(2/5)

- A simple baseline: Generalizable human NeRF from monocular videos
- Key design
  - Canonical space
  - Bidirectional LBS

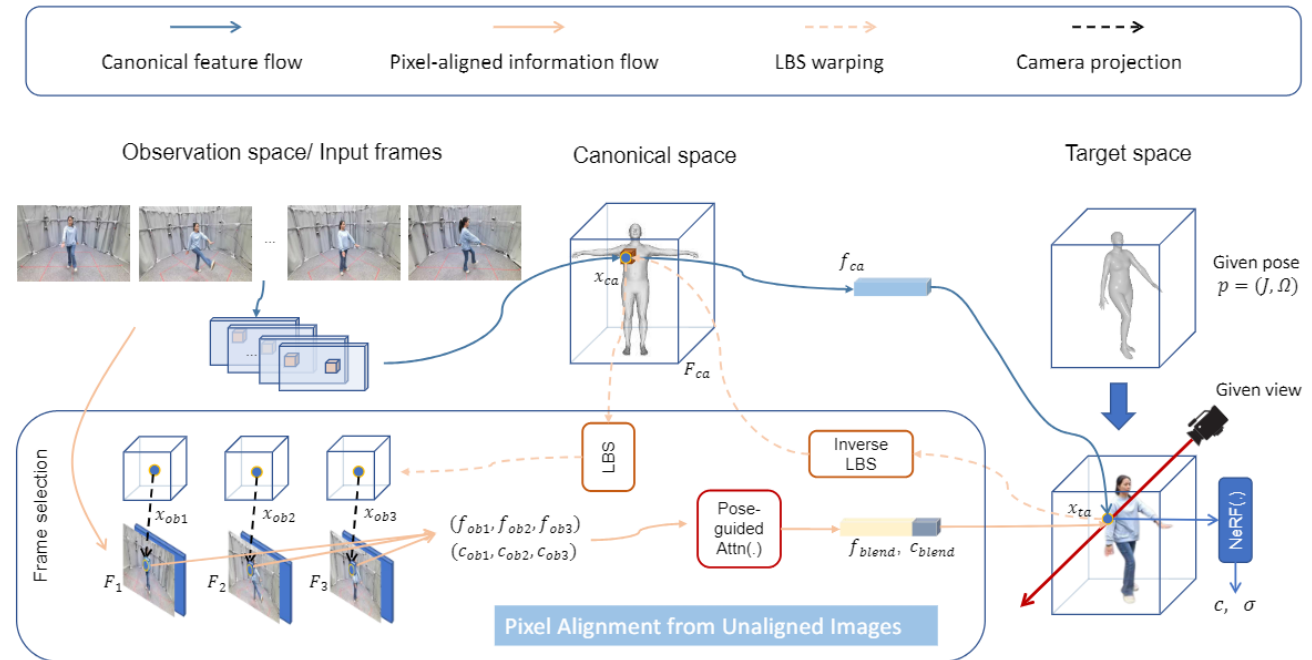


Figure 31: Method for 3D Human Animation





# 3D Human Animation(3/5)

- Fair comparison with the same settings

Table 8: Fair comparison with the same settings

|                | Methods            | Reported in paper |                 |                    | Unified evaluation |                 |                    |
|----------------|--------------------|-------------------|-----------------|--------------------|--------------------|-----------------|--------------------|
|                |                    | PSNR $\uparrow$   | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$    | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| Scene-specific | NeuralBody [10]    | 28.10             | 0.9440          | -                  | <b>27.80</b>       | <b>0.9400</b>   | 0.1050             |
|                | AniNeRF [197]      | 27.10             | 0.9490          | -                  | 24.10              | 0.8820          | 0.1870             |
|                | HumanNeRF [47]     | <b>30.20</b>      | <b>0.9680</b>   | <b>0.0317</b>      | 25.86              | 0.9245          | 0.0728             |
|                | UV-Volume [196]    | 27.95             | 0.9346          | 0.0720             | 26.45              | 0.9282          | <b>0.0726</b>      |
| Generalizable  | NHP [200]          | 24.80             | 0.9050          | -                  | 24.68              | 0.9034          | 0.1706             |
|                | GP-NeRF [199]      | <b>26.00</b>      | <b>0.9210</b>   | -                  | <b>26.66</b>       | <b>0.9263</b>   | 0.1256             |
|                | KeypointNeRF [198] | 25.03             | 0.8969          | -                  | 26.01              | 0.9159          | <b>0.1041</b>      |





# 3D Human Animation(4/5)

- Scene-specific VS generalizable

Table 9: Scene-specific VS generalizable

| Methods        | 15 mins finetune |                 |                    | 1 hour finetune |                 |                    | 20 hours finetune |                 |                    |
|----------------|------------------|-----------------|--------------------|-----------------|-----------------|--------------------|-------------------|-----------------|--------------------|
|                | PSNR $\uparrow$  | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ | PSNR $\uparrow$   | SSIM $\uparrow$ | LPIPS $\downarrow$ |
| GP-NeRF [199]  | <b>27.46</b>     | <b>0.9275</b>   | <b>0.1464</b>      | <b>27.94</b>    | <b>0.9330</b>   | 0.1390             | <b>28.80</b>      | <b>0.9408</b>   | 0.1290             |
| HumanNeRF [47] | 22.22            | 0.8549          | 0.1772             | 23.26           | 0.8779          | <b>0.1285</b>      | 25.36             | 0.9156          | <b>0.0807</b>      |

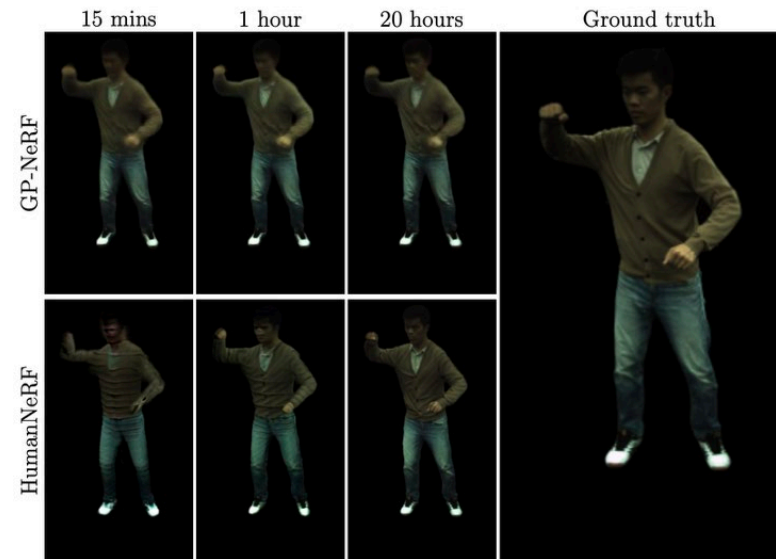


Figure 31: Scene-specific VS generalizable







# 3D Human Animation(5/5)

- A simple baseline: Generalizable human NeRF from monocular videos

Table 10: The performance of our built baseline

| Train set   | Test set    | PSNR $\uparrow$ | SSIM $\uparrow$ | LPIPS $\downarrow$ |
|-------------|-------------|-----------------|-----------------|--------------------|
| ZJU-MoCap 7 | ZJU-MoCap 3 | <b>22.87</b>    | <b>0.8647</b>   | <b>0.2110</b>      |
| HuMMan      | ZJU-MoCap 3 | 22.73           | 0.8573          | 0.2290             |
| ZJU-MoCap 7 | HuMMan Eval | 18.09           | 0.8715          | 0.2159             |
| HuMMan      | HuMMan Eval | <b>21.06</b>    | <b>0.8882</b>   | <b>0.2010</b>      |





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# Conclusions

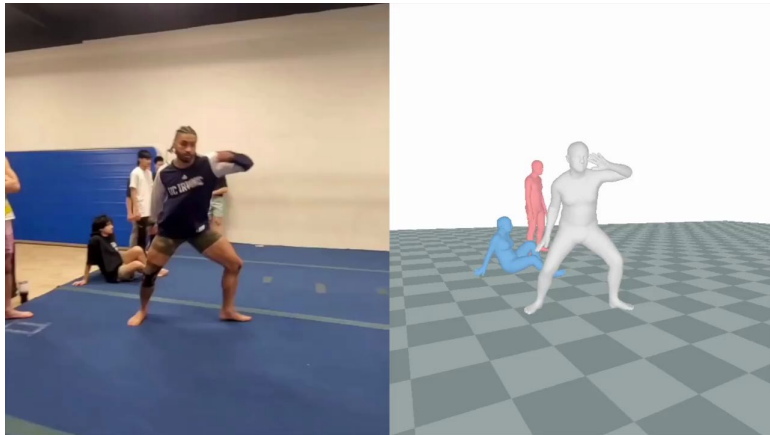


Figure 32: Capturing human motions from videos

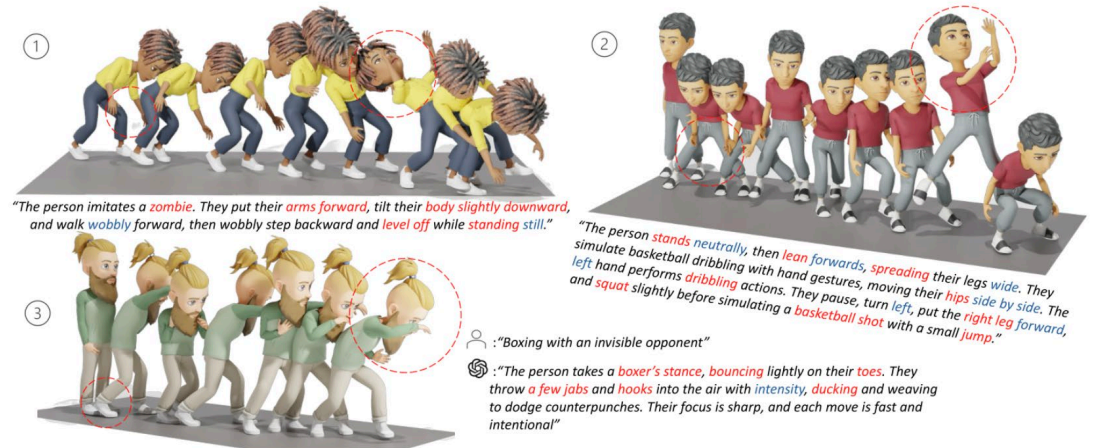


Figure 33: diverse real-world conditions





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# Q&A

Thank You!

